another end to a weight passes over the pulley tightly. Note down the initial reading of the pointer. Keep the set up for a week.

Observation :

The stem tip grows in length, the pulley moves, and the pointer slide over the graduated arc. The distance travelled by the pointer is noted down.

Inference :

Actual growth in length is calculated with help of this formula.

$$\text{Actual growth in length} = \frac{\text{Distance travelled by the pointer} \times \text{radius of the pulley}}{\text{Length of the pointer}}$$

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